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Roll No.....

MI-305 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Engineering Materials
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. What are different types of Engineering Materials? Enumerate any two properties from mining engineering point of view.
2. Briefly explain Iron-Carbon phase diagram. Write down all invariant reactions and factors affecting phase transformation.
3. Write down the composition, properties and uses of following types of Alloy steel in brief:
 - i) Tungsten Steel
 - ii) Manganese Steel
4. a) Differentiate between elastic and plastic deformation.
b) What is creep. Discuss its importance.
5. Write down the composition and properties of following engineering materials:
 - a) Portland cement
 - b) Quick setting cement
 - c) Shotcrete
6. What is hardening of steels? What factors are taken into account while deciding hardening of steel?

OR

What is a Resin Capsule? Explain its importance as a material used in Mining engineering.

7. With the help of sketches explain the various types of wire ropes and constructions of wire ropes.

OR

What is annealing. Explain diffusion annealing.

8. Explain the following : (any two)
 - i) Quenching
 - ii) Nitriding
 - iii) Tempering
 - iv) Normalizing
